

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Gaskets

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Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	22-06-22	First Issue for Clients Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- MPP-SPC-014

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Class Specification
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping

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Abbreviations

Abbreviations	Meaning
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
CAF	Compressed Asbestos Fibre
CNAF	Compressed None Asbestos Fibre
CS	Carbon Steel
DEP	Design and Engineering Practice
FEED	Front End Engineering Design
MESC	Material and Equipment Standard and code
MMMF	Man Made Mineral Fiber
MSS	Manufacturers Standardisation Society
MTO	Material Take Off
REF	Reference
SS	Stainless Steel

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

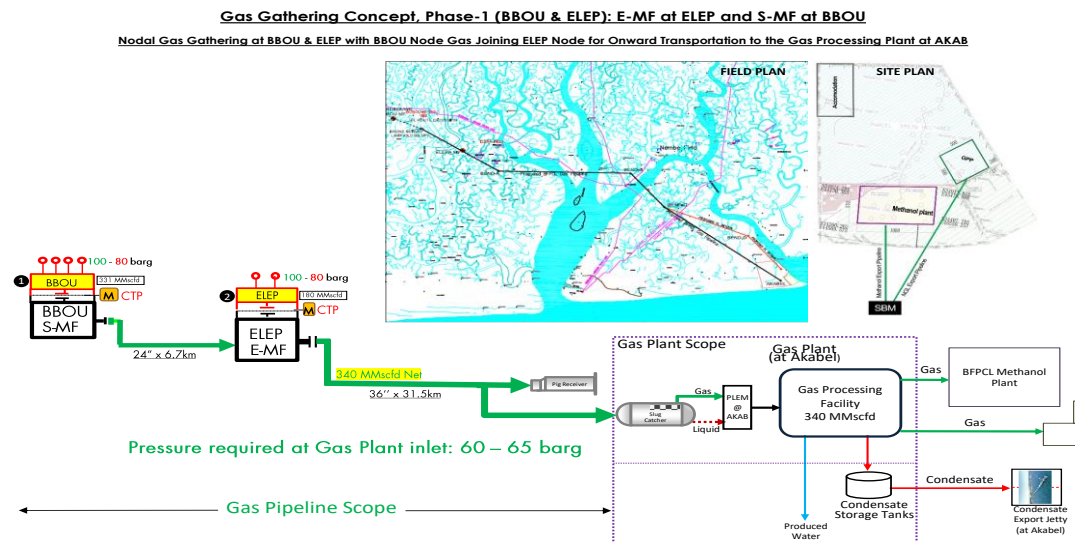


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification covers the minimum technical requirement for design, manufacturing, handling and supply of non-metallic flat gaskets, spiral wound and Ring type gaskets that shall be used for the Pipeline and associated piping system of the BFPCL Gas Gathering Project-Phase 1.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.24	Piping – Engineering and Layout Requirements
DEP 30.10.02.13	Non-metallic Materials – Selection and Application
MESC Group 85	Gasket and packing
ASME B16.20	Metallic Gaskets for Pipe Flanges - Ring-Joints, Spiral-Wound, and Jacketed
ASME B16.21	Non-metallic Flat Gaskets for Pipe Flanges
ASME B16.47	Large Diameter Steel Flanges NPS 26 Through NPS 60
ASME B16.5	Pipe Flanges and Flanged Fittings NPS ½ Through NPS 24
ASTM F 36	Standard Specification for Test Method for Compressibility and Recovery of Gasket Materials
ASTM F 104	Standard Specification for Classification Systems for Non-metallic Gasket Materials
ASME B31.3	Process Piping
MSS SP 6	Standard Finishes for Contact Faces of Pipe Flanges and Connecting-Ends Flanges of Valves and Fittings
EN 10204	Metallic products - Types of inspection documents
ASTM A 262	Standard Practices for Detecting Susceptibility to Intergranular Attack in Austenitic Stainless Steels
API 6FB	Specification for Fire Test for End Connections

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract

- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Description of Gasket Materials

3.1. General

Gaskets shall comply with this specification and referenced ASME and ASTM standards. The gasket type, rating, and materials of construction shall be in accordance with the MTO referenced in the Requisition.

Asbestos containing gaskets are prohibited. Gaskets shall have proven resistance to the fluids conveyed in the associated piping. It is the Bidder's responsibility to seek additional information on the conveyed fluids, if required.

The Vendor shall provide a written guarantee that the selected gaskets are chemically inert to the listed materials at the stated design conditions.

Gaskets shall suit the following flange design standards: sizes up to and including 24" – ASME B16.5; Larger sizes – ASME B16.47 Series A.

3.2. Cut-out Flat Gaskets

Flat gaskets shall be based on the following requirements:

- Flat gaskets shall be supplied in accordance with the technical requirements of ASME B16.21
- Compressed asbestos fibre (CAF) shall not be used as flat sheet gasket. Alternative to CAF shall be foil or expanded metal reinforced graphite gaskets.
- Manmade mineral fibres (MMMFs) shall not be used as flat sheet gasket
- Synthetic rubber gaskets shall preferably be made from neoprene fabric or equivalent.
- PTFE or PTFE based materials are not acceptable. Only fiber-based materials or flexible graphite materials are acceptable.
- The material data sheet shall include the following mechanical characteristics for the two referenced thicknesses 2 mm and 3 mm in the Piping material classes.
- The tensile strength (according to ASTM F 152) of the gasket material shall be at least 16 N/mm².
- The compressibility of the gasket material measured according to ASTM F 36A shall be at least 8%.
- The elastic recovery of the gasket material according to ASTM F 36A shall be at least 55%.
- The relaxation stress (according to DIN 52913, short term test) of the gasket material shall be at least 30 N/mm² (ideally, a minimum of 36 N/mm²), and the thickness decrease shall not exceed 15% under a constant compression stress of 50 N/mm² @ 300°C.
- The thickness increase under immersion test in ASTM oil no. 3 @ 150°C for 5 hours shall

not exceed 15%. Additionally, the thickness increase under immersion test in fuel ASTM A @ 20°C for 5 hours shall not exceed 3%, and the thickness increase under immersion test in fuel ASTM B @ 20°C for 5 hours shall not exceed 8%.

- The inserts in flexible graphite materials, gaskets, or sheets to be cut to size, must be in one piece without any weld. Cutting of flexible graphite sheets or reinforced fibre-based materials shall be carried out by water jet means followed by an adequate heat treatment or by mechanical tools and procedure approved by the gasket Manufacturer; other mechanical tools are prohibited.
- The thickness of the gasket material shall be strictly identical to the thickness of the equivalent CAF gasket as shown on the relevant piping material classes (i.e., 2 or 3 mm, depending on diameter). Gas permeability (according to DIN 3535): = 0.5 ml/mm.
- Gasket faces shall receive an anti-sticking coating on both sides which shall be approved by the Owner before application.
- If reference is made to the ASTM F 104 for a proposed gasket material, the classification shall be F702100 E03 M7 together with the additional requirements listed in the paragraphs above of this section (which are not covered in the standard ASTM F 104).

3.3. Spiral-wound Gaskets

The gaskets shall be in accordance with the standard ASME B16.20 and their spiral-wound portion shall be made from type 316 stainless steel (or higher grade, depending on the relevant piping material specification) strip.

The following requirements shall apply:

The requisitions shall indicate the nature of the packing of the filling.

- GRAPHITE based fillers shall incorporate an adequate corrosion inhibitor to prevent possible galvanic corrosion.
- The gaskets shall have an external centring ring and an inner ring. The two rings are made of either an epoxy-polyester coated carbon steel or of stainless steel (i.e., made of the same metallic material as for the stainless-steel strip spires). In case of epoxy-polyester coating, the painting quality shall satisfy the following criteria:
 - The finished thickness shall be between 60 µ and 100 µ
 - There shall be one coat only
- Testing shall be made on one test coupon per batch for the following:
 - Adhesion (ASTM D 3359)
 - Impact (ASTM G 14)
 - Corrosion (Salt spray test per ASTM B 117).

3.4. Metal Ring Joints

The Gasket shall be of the octagonal ring type

Gaskets shall have to following maximum Brinell hardness:

- Soft Iron: 90
- Alloy steel: 130
- 18/8 Stainless Steel: 135

4. Dimensions

4.1. Cut out Flat Gaskets

4.1.1. Synthetic Rubber Gaskets

The gaskets shall be 3 mm thick for diameters up to 6" and 5 mm thick for diameters 8" and over.

4.1.2. Other Flat Gaskets

Materials (approved by the Owner) shall have dimensions (including the gaskets thickness) identical to those of the CAF gaskets which they are intended to replace.

4.2. Spiral-wound Gaskets

The dimensions of spiral-wound gaskets used on raised contact faces shall be in accordance with the standard ASME B16.20.

4.3. Metal Ring Joints

Ring joints may be either oval or octagonal. The dimensions of ring joints shall be in accordance with the standard ASME B16.20, and shall be specified to match with the relevant flanges standard.

4.4. Special Gaskets

For all other types of gaskets, or gaskets of dimensions greater than the limits defined by the reference standards, the Manufacturer shall submit a dimensioned document to the Owner for approval.

5. Quality Control

The inspection and tests necessary to confirm that the products meet the requirements of the standards, specifications, and requisitions, shall be carried out in the Manufacturer's plant by personnel of the plant. This specially qualified personnel shall be independent of the production department of the plant.

Products shall be guaranteed by an inspection certificate (conformity, material, etc.) that must accompany the supply. If the Manufacturer does not have the necessary means of examination, these tests may be carried out by an Inspection Agency and at a place approved by the Owner.

6. Manufacturer's Dossier

6.1. Certificate of Conformity

The Manufacturer shall submit certificates of conformity to the standards specified by the requisitions for cut-out gaskets, spiral-wound gaskets, and metal ring joints.

6.2. Material Certificates

The Manufacturer shall submit material certificates in accordance with the requirements of the reference codes and standards. For non-asbestos gaskets (i.e., either flat gaskets or spiral wound gaskets), certificates shall be in accordance with EN 10204 type 3.1.B.

7. Preparation for shipment

7.1. Marking

7.1.1. Cut-out Flat Gaskets

A label grouping several gaskets of the same type shall indicate:

- The Manufacturer's name and trademark
- The diameter
- The rating
- The quality reference.

7.1.2. Spiralwound Gaskets

In addition to the marking required by the standard ASME B16.20, the following information shall be marked by cold stamping on the centring ring of each spiral-wound gasket:

- The MANUFACTURER's name and trademark
- The diameter
- The rating.

7.1.3. Metal Ring Joints

In addition to the marking required by the standard ASME B16.20, the outside surface of each ring shall include the following information which shall be marked by cold stamping: T

- The Manufacturer's name and trademark
- The diameter
- The rating
- The ring number, followed by the identifying reference of the material.

7.2. Protection

Ring joint gaskets other than those made of stainless steel shall be covered by grease impregnated protecting strips.

Small diameter gaskets shall be grouped in cardboard boxes or plastic pouches and large diameter gaskets in rigid packing cases.

7.3. Colour Coding

Combination gaskets (spiral wound gaskets) shall be coloured in accordance with ASME B16.20

Paints, coating, or plating shall not contain cadmium, zinc or any other detrimental metal

7.4. Packing

The gaskets must be packed in sea-packing, which must be tight and meet the requirements of all stages of transport (rail, road, air, etc.).